



Dhaivat Joshi

| Embedded Firmware Engineer | PCB Designer |

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👤 PROFILE

Master's student in Electrical Engineering and Information Technology at Otto-von-Guericke University Magdeburg with hands-on experience in embedded systems, Linux-based development, PCB design, and IoT applications. Skilled in error analysis, system validation, GUI prototyping, and Hardware prototyping.

🏢 PROFESSIONAL EXPERIENCE

05/2025 – 07/2025
Delhi, India

PCB Designer | Embedded Firmware Developer ✎
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- Developing ESP32-based systems with FreeRTOS for multi-sensor protocol to USB Serial communication.
- Designing custom PCBs, including an Ethernet splitter for drone applications, using KiCad 9.
- Programmed and validated Linux/RTOS-based embedded systems with Ubuntu for testing and evaluation.
- Conducted software verification and bug analysis for multi-sensor applications, reporting results in Git-based workflows.

06/2021 – 10/2023
Anand, India

Embedded Firmware Developer | PCB Designer
InSignEx

- Embedded Systems & IoT Expertise: Developed and optimized firmware for microcontrollers (PIC18, ESP8266EX, AVR Atmega328p), and integrated IoT solutions using protocols like MQTT, SPI, I2C, UART, and MODBUS.
- Hardware Design: Designed and optimized schematics and PCB layouts in KiCad for various projects, including a Hydroponic System and a voice-controlled scrolling display, completing six hardware projects in total.
- Full-Stack Development: Developed user interfaces for the Gh5001 Green House Monitoring System app, utilizing MQTT, Thingspeak, and Firebase for data-driven applications, and developed three custom Android apps with Kodular.
- Programming Skills: Proficient in C, C++, and Python, with experience in RTOS development and creating three C libraries to enhance system functionality.
- Project Management: Managed project data and development using tools like Visual Studio, Arduino IDE, and Excel with Python, streamlining workflows for multiple hardware and software projects.

🎓 EDUCATION

10/2023 – present
Magdeburg, Germany

M.TECH ELECTRICAL ENGINEERING AND INFORMATION TECHNOLOGY
Otto-von-Guericke Universität Magdeburg ✎
Field of Focus: Electrical, Embedded System, Control System, Power Electronics

07/2017 – 04/2021
Anand, India

B.TECH ELECTRONICS AND COMMUNICATION ENGINEERING
Charotar University of Science and Technology ✎
Field of Focus: Embedded Systems, Fundamentals of Electronics and Power Electronics
Final grade: 8.01 CGPA or Equivalent to 2.0 in Germany

PROJECTS

IOT Hydroponic System

InSignEx

- Designed and validated a fully automated hydroponic system with IOT control via ESP8266, managing pH and humidity sensors, pumps, and feeders on a 12V DC supply, Designed custom schematics and PCB layout using KiCad.
- Integrated an RTC for precise relay timing, enabling automated Schedule and Repeat modes, performed hardware verification and software validation to ensure reliable long-term operation.
- Created and tested an Android app for seamless system configuration over MQTT using JSON data, supported by RTOS-based firmware for efficient control and real-time monitoring.

Electronic steering wheel with pneumatic shifting system

OJASWAT MOTORSPORT

- Designed and developed a smart steering wheel for a Formula Student car to control critical systems.
- Integrated gear shifting, radiator fan control, and real-time sensor data display using dedicated microcontrollers.
- Implemented CAN BUS communication for efficient inter-controller data sharing and enhanced driver feedback.

RP2040 Macropad

Personal Project | Sponsored by JLCPCB & Seeed Studio

- Designed a custom RP2040-based USB macropad with mechanical keys, a rotary encoder, and RGB lighting, including the schematic and PCB layout in KiCad. Configured QMK firmware for programmable keymaps, macros, and media controls, with PCB fabrication and components sponsored by JLCPCB and Seeed Studio.

USBASP AVR Programmer

B.Tech Project

- A USB in-circuit programmer for Atmel AVR controllers that allows you to install new bootloaders and firmware on popular AVR microcontrollers.

TECHNICAL SKILLS

Software - Proficient

GIT/Github, GitLab, KiCad, EasyEDA, Proteus, Microsoft Office (Word | Excel | Power Point), Arduino IDE, STM32Cube IDE, PlantUML, Jenkins, CI/CD

Communication protocols - Competent

IOT, MQTT, SPI, I2C, UART, ESPNOW, MODBUS, CANBUS

Programming language - Competent

C, C++, Embedded C, ROS, Python, PHP, HTML(basics)

Floor skills - Expert

Rapid prototyping, micro soldering, circuit debugging, PCB production

Microcontroller Board - Competent

STM32, ESP8266ex, ESP32, PIC18, ATtiny85, Arduino UNO/ Mega, Raspberry pi

OS

Linux(Ubuntu), windows, FreeRTOS

LANGUAGES

German — Basic

Goethe Zertifikat A2 (currently improving)

English — Fluent

EXTRACURRICULAR ACTIVITIES

02/2019 – 03/2021

Anand, India

Electrical Team Lead

Ojaswat Motorsport - University's Formula Student Team

- Led an electrical Team, converting a racing bike circuit to suit a Formula student car application.